ME 132: Dynamic Systems and Feedback

University of California, Berkeley
Mechanical Engineering Department
Negar Mehr

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Instructor's Office Hours: M, 11:30 am - 1 pm | Class Hours: M/W/F, 10 - 11 am

Course Description

This course provides an introduction to the analysis and design of feedback control systems. Topics include the role of feedback in achieving precision and robustness, the representation of dynamic systems using signals and block diagrams, and the fundamental components of control systems, including plants, sensors, actuators, references, disturbances, and controllers. Students will examine the distinction between open-loop and closed-loop architectures, evaluate system performance with respect to measures such as settling time, overshoot, stability, and disturbance rejection, and consider robustness to modeling uncertainty as a key design criterion. The course further addresses the complete control system design process such as the selection of regulated variables, sensor and actuator placement, plant modeling and system identification, controller synthesis, and validation through simulation. We will extensively use Matlab and Simulink for the modeling and simulation of dynamic systems, with applications drawn from a broad range of engineering domains, including robotics, process control, and semiconductor manufacturing.

Course Materials

There is no required text for the course. Notes will be provided for *most* of the topics covered. It is strongly suggested that students attend lecture and take good notes. The course material will partly follow the following references:

- *Feedback Systems: An Introduction for Scientists and Engineers*, K. Astrom and R. Murray, Princeton University Press, Princeton, NJ, 2008 (Availabe on bCourses as a pdf).
- *Feedback Control of Dynamic Systems*, G. F. Franklin, J. David Powell, and A. Emami-Naeini, Prentice-Hall, 6th edition, 2009.

While the above textbooks will be useful for the course, the material covered will be significantly smaller than the union of these books.

Prerequisites

Students are expected to have taken Math 53, Math 54, Physics 7A, Physics 7B, and E7. Some previous exposure to MatLab will be valuable. ME 104 or equivalent can be taken concurrently.

Staff

- *Instructor*:
  - Negar Mehr (`negar@berkeley.edu`)
- *GSI*:
  - Seoyeon Choi (`seoyeon99@berkeley.edu`)
  - Ryan Park (`ryanjpark03@berkeley.edu`)
- *Reader*:
  - Evan Griffith (`evangriffith@berkeley.edu`)
  - Nathan Jew (`nathan.jew@berkeley.edu`)

Office Hours

- Dr. Mehr's office hours:
  - Date and time: Mondays 11:30-13
  - Location: 5126 Etcheverry Hall
- Seoyeon's office hours:
  - Date and Time: TBD,
  - Location: TBD
- Ryan's office hours:
  - Date and Time: TBD,
  - Location: TBD

Communication

Course announcements and important updates will be posted on bCourses, and all students are expected to have regular access to the platform and check it frequently. We also encourage students to use Ed for course-related discussions and questions as this allows both the instructional team and other students to participate in the conversation. Students are especially encouraged to contact the GSIs through Ed rather than by email whenever possible so that questions and requests do not get lost in individual inboxes and can be addressed more efficiently.

Course Structure

The course will consist of in-person lectures and lab (discussion) sections, regular homework assignments, two in-class midterm exams, and a final exam.

Exam Dates

The following are exam dates:

- Midterm 1: Friday 10/9/2026, *in class*
- Midterm 2: Friday 11/13/2026, *in class*
- Final: Monday 12/14/2026, 8-11 am, location: TBD

Homework

Homework is *very important*. It is the best way to learn the theory material. Homework will be posted on bCourses on Fridays and will be due by **11:59 p.m. sharp on Sunday of the following week**.

We will be using Gradescope to assign and submit homework. You will receive a link via your Berkeley email. Email GSIs if you have any issues getting to the ME 132 Gradescope website. For help scanning and submitting homework, please see this link.

Matlab Proficiency

Please download and install a free copy of Matlab and Simulink from Software Central. We assume everyone has taken E7 or a comparable programming class. If you are unfamiliar with the basics of Matlab, make an appointment to meet with a GSI as soon as possible.

Introduction to Simulink Sessions

Introduction to Simulink sessions will be offered during the first two weeks of instruction during regular lab times. You must attend at least one session if you have not used Simulink.

Lab Sections

There will be four lab sections each week, led by the GSIs. Labs will function like discussion sections during which the GSIs will cover the course material and questions. *Students are expected to attend the lab section for which they are officially registered*.

Course Policies

Grading Policy

The deliverables for the course will consist of homework assignments, midterms, and a final exam. The weights for the deliverables will be distributed as follows:

- Homework: **10%**
- Midterm 1: **20%**
- Midterm 2: **20%**
- Final: **50%**

Late Homework Policy

Late homework will not be accepted under any circumstance including internet issues. This is because it isn't fair to other students and because the logistics of a 200-student course are formidable. We understand that sometimes you will miss turning in homework, or will be too busy with other classes. So *we will simply drop your two lowest homework scores from grade calculations* to account for absences, birthdays, bad days, and other realities of life. This means you could miss turning in homework two times without penalty. Homework solutions will be posted on bCourses after the due date.

Academic Integrity and Honesty

We have a *zero tolerance* policy towards cheating. We will make serious efforts to proctor exams closely. Cheating is disrespectful to the vast majority of other students who work hard and honestly represent their own work. The risk just isn't worth it to get a slightly better grade in one course out of many you will take.

Schedule and Weekly Learning Goals

The schedule is tentative and subject to change. *Note that the first lecture will take place on Wednesday, August 26, 2026.* The learning goals below should be viewed as the key concepts you should grasp after each week, and also as a study guide.

Week 1, 08/24 - 08/28: The power of feedback; terminology, structure of control systems, block diagrams, major concepts.

Week 2, 08/31 - 09/04: Cruise control example; signals, systems, models, controllers.

Week 3, 09/07 - 09/11: First order systems; free response, step response, frequency response, transfer functions; control.

Week 4, 09/14 - 09/18: Second order systems; PID control; anti-wind-up strategies.

Week 5, 09/21 - 09/25: General LTI systems; transfer functions; stability; frequency response.

Week 6, 09/28 - 10/02: Interconnected systems; root-locus; Nyquist theorem; delays.

Week 7, 10/05 - 10/09: Stability robustness and robustness margins.

Week 8, 10/12 - 10/16: Linear algebra review; introduction to state-space.

Week 9, 10/19 - 10/23: Realizations; general solution.

Week 10, 10/26 - 10/30: State-feedback; pole-placement; observers; separation principle.

Week 11, 11/02 - 11/06: Nonlinear systems; equilibrium points; linearization.

Week 12, 11/09 - 11/13: Feedback linearization; Gain-scheduling.

Week 13, 11/16 - 11/20: System identification; estimation; model predictive control; adaptation.

Week 14, 11/23 - 11/27: *Thanksgiving Break (the lecture on Monday 11/23 will be a recorded lecture).*

Week 15, 11/30 - 12/04: New frontiers.

Inclusion

We are committed to creating an environment welcoming of all students where everyone can fulfill their potential for learning. To do so, we commit to support a diversity of perspectives and experiences and respect each others' identities and backgrounds (including race/ethnicity, nationality, gender identity, socioeconomic class, sexual orientation, language, religion, ability, etc.). To help accomplish this:

- If you feel that your performance in the class is being impacted by a lack of inclusion, please contact the instructor, an academic advisor, or the departmental Faculty Equity Advisor. An anonymous feedback form is also available here.
- If you feel that your performance in the class is being impacted by your experiences outside of class (e.g., family matters, current events), please don't hesitate to come and talk with the instructor or academic advisors in Engineering Student Services. We want to be a resource for you.
- There is no tolerance for sexual harassment or violence. If your behavior harms another person in this class, you may be removed from the class or the University either temporarily or permanently.
- If you have a name and/or pronouns that differ from your legal name, designate a preferred name for use in the classroom here.
- As a participant in this class, recognize that you can be proactive about making other students feel included and respected.

Berkeley Honor Code

Everyone in this class is expected to adhere to this code: "As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others".

Student Conduct

Ethical conduct is of utmost importance in your education and career. The instructors, the College of Engineering, and U.C. Berkeley are responsible for supporting you by enforcing all students' compliance with the Code of Student Conduct and the policies listed in the CoE Student Guide. The Center for Student Conduct is your central source for guidance in these matters.

Accommodation Policy

We honor and respect the diversity in our student body and thus are committed to ensuring you have the resources you need to succeed in our class. If you need accommodations that provide equitable access (e.g. religious observance, physical or mental health concerns, insufficient resources, etc.), please check https://diversity.berkeley.edu/ and, if needed, discuss your specific case with the instructor.

Resources

- **For disability accommodations:** The Disabled Student's Program (DSP 260 César Chávez Student Center 4250; 510-642-0518) serves students with disabilities of all kinds, including temporary disabilities (e.g., broken arm impacting ability to write). Services are individually designed and based on the specific needs of each student as identified by DSP's specialists. If you have already been approved for accommodations through DSP, please know that DSP is ready to quickly adjust your accommodations if your situation changes.
- **For mental wellbeing**: Counseling and Psychological Services (CAPS) is available as part of University Health Services (the Tang Center). Services are offered at many locations, including on-site in the College of Engineering. CAPS services are available to all students, regardless of insurance, and initial visits do not cost anything. CAPS has expanded allowing students to receive help immediately with same-day counseling (510-642-9494), online resources, and a 24/7 counseling line at (855) 817-5667. Short-term help is also available from the Alameda County Crisis hotline: 800-309-2131. If you, or someone you know, is experiencing an emergency that puts their health at risk, please call 911.
- **For basic needs (food, shelter, etc):** The Basic Needs Center provides housing, food, transportation support, among other support needed to thrive at UC Berkeley. Specifically, the UC Berkeley Food Pantry (68 Martin Luther King Student Union) aims to reduce food insecurity among students, especially the lack of nutritious food. Students can visit the pantry as many times as they need and take as much as they need while being mindful that it is a shared resource. The pantry operates on a self-assessed need basis; there are no eligibility requirements. The pantry is not for students and staff who need supplemental snacking food, but rather, core food support.

Modifications to the Syllabus

The instructor reserves the sole right to modify any and all parts of this syllabus throughout the semester. All modifications will be made solely in the interest of time scheduling, accurately measuring the students' success, and improving the students' educational outcomes.